

Test and Evaluation Department

Ballistic Testing Facility

Test Report

Ballistic Resistance Test

55.80mm BR-5 (NS) Ballistic Glass

Batch No.70745

EN 1063 Ballistic Standard BR5- (NS)

(5.56x45 mm SS109)

40.60mm BR-4 (NS) Ballistic Glass

Batch No.70745

EN 1063 Ballistic Standard BR4- (NS)

(.44 Rem. Magnum)

UNIGLASS /KSA

JODDB/TEST/BTF/TR/1029



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Approval Test Report

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Batch No.70745
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Batch No.70745
EN 1063 Ballistic Standard BR4- (NS)
(.44 Rem. Magnum)

UNIGLASS /KSA

Compiled by

Head of Weapons, Ammunition & Armor Testing
Eng. Issa Abed AL Rahman Rawashdeh

Approved by

Head of Test & Evaluation Department
Dr. Eng. Riyadh Ali Ratrout

Signature

Date



9 January 2023

1. Test Data

Requester	JODDB / Sales
Contractor	UNIGLASS ,Al Hara, Al Mashael, Riyadh 14325, Saudi Arabia
References	- Test and Evaluation Request No. 1029 dated 02/02/2022 - Test Proposal No .JODDB/TEST/BTF/TP/1029 Dated 02/02/2022 (See Appendix A)
Test Samples	Test Sample No.1: BR-5 Test Samples: 55.80mm BR-5 (NS) Ballistic Glass / Batch No.70745 - Test Piece No.1: 55.80 mm Ballistic Glass/ Size 500x500mm/ Batch No.70745/ Weight 33.335 kg - Test Piece No.2: 55.73 mm Ballistic Glass/ Size 500x500mm/ Batch No.70745/ Weight 33.625 kg - Test Piece No.3: 55.85 mm Ballistic Glass/ Size 500x500mm/ Batch No.70745/ Weight 33.380 kg Test Sample No.2: BR-4 Test Samples: 40.60mm BR-4 (NS) Ballistic Glass / Batch No.70745 - Test Piece No.1: 40.62 mm Ballistic Glass/ Size 500x500mm/ Batch No.70745/ Weight 24.445 kg - Test Piece No.2: 40.61 mm Ballistic Glass/ Size 500x500mm/ Batch No.70745/ Weight 24.360 kg - Test Piece No.3: 40.67 mm Ballistic Glass/ Size 500x500mm/ Batch No.70745/ Weight 24.360 kg (See Photos No. 1)
Manufacturer	UNIGLASS /KSA
Test Type	Ballistic Resistance Test
Test Standard	EN 1063 Ballistic Standard level BR5-(NS) & BR4-(NS)
Test Site	Ballistic Testing Facility (JODDB)
Samples Received Date	27 th December 2022
Test Temperature	22 °C
Test Weapon	- NATO Universal Ballistic Breech with standard barrel: 5.56x45mm SN:3715 - NATO Universal Ballistic Breech with standard barrel: .44 Rem Mag SN:5048
Type of Bullets	5.56x45 mm SS109 Full Steel Jacket, Pointed Bullet lead-soft core steel penetrator, Bullet Mass (4.0 ± 0.1). Batch. No. 605-FNB-08 0.44 Rem. Magnum, Full Copper Alloy Jacket Flat Nose, Soft Core (Lead), Bullet Mass (15.6±0.1 g), Batch No Lot No. 227-1602
Test Equipment	- Doppler Radar SN.3503-13-270-011 (± 0.4 m/s) - Digital caliper SN. 112167 (± 0.07 mm) - Temp .Hum. Meter SN.95941255 (± 0.3 °C) - Distance laser Measuring Device (Lacia) SN. 1030969032 (±2 mm) - Precision Balance SN. KADDB-011 (±0.062 kg)
Test Conducted By	- Eng. Ali Sardyah - Eng. Sameer Obeidat - Eng. Mohannad Akilan - Eng. Ramiz Al-Tawal - Senior Technician: Hasan Aldebis - Senior Technician : Jalal Gholilat - Technician: Moath Anagreh.
Test Date	03 January 2023

Note: The results contained in this report are only valid for detailed above and the report shall not be reproduced except in full without approval of the laboratory.

2. Test Results

55.80mm BR-5 (NS) Ballistic Glass (NS)/ Batch No.70745				
Test Sample No.1	Number of Shots	Bullet Velocity (m/s)		
		1 st	2 nd	3 rd
Test Piece No.1 55.80 mm Ballistic Glass Batch No.70745 Weight 33.335 kg	Three Shots 5.56x45 mm SS109 Full Steel Jacket, Pointed Bullet lead-soft core steel penetrator, Bullet Mass (4.0 ± 0.1) Test Range: 10 ±0.5 m. Bullet Velocity (Doppler): 950±10 m/s Striking Distance: 120±10 mm	958.3	949.4	957.3
Status : (P / NS / S) P– Perforation / NS - No Perforation & No Splinters / S - No Perforation with Splinters		NS	NS	NS
Test Results		No Perforation & No Splinters (See Photo No.2)		
Test Piece No.2 55.73 mm Ballistic Glass Batch No.70745 Weight 33.625 kg	Three Shots 5.56x45 mm SS109 Full Steel Jacket, Pointed Bullet lead-soft core steel penetrator, Bullet Mass (4.0 ± 0.1) Test Range: 10 ±0.5 m. Bullet Velocity (Doppler): 950±10 m/s Striking Distance: 120±10 mm	943.1	953.8	954.2
Status : (P / NS / S) P – Perforation / NS - No Perforation & No Splinters / S - No Perforation with Splinters		NS	NS	NS
Test Results		No Perforation & No Splinters (See Photo No.2)		
Test Piece No.B3 55.85 mm Ballistic Glass Batch No.70745 Weight 33.380 kg	Three Shots 5.56x45 mm SS109 Full Steel Jacket, Pointed Bullet lead-soft core steel penetrator, Bullet Mass (4.0 ± 0.1) Test Range: 10 ±0.5 m. Bullet Velocity (Doppler): 950±10 m/s Striking Distance: 120±10 mm	952.7	948.3	955.0
Status : (P / NS / S) P – Perforation / NS - No Perforation & No Splinters / S - No Perforation with Splinters		NS	NS	NS
Test Results		No Perforation & No Splinters (See Photo No.2)		
Status(Comply - See Applied Statement of Conformity and Decision Rule - Page 6)				
EN1063 Performance Requirements P - Perforation: Piercing of a test piece by a bullet or by bullet fragments, and/or creation of an opening from the attack face to the rear face. S - No Perforation with Splinters: No perforation of the test piece by the bullet or parts of the bullet. But with perforation of the witness foil by glass splinters from the protected face. NS – No Perforation & No Splinters: No perforation of the test piece by the bullet or parts of the bullet. And no perforation of the witness foil by glass splinters from rear face.				

40.60mm BR-4 (NS) Ballistic Glass / Batch No.70745				
	Number of Shots	Bullet Velocity (m/s)		
		1 st	2 nd	3 rd
Test Piece No.1 40.62 mm Ballistic Glass Batch No.70745 Weight 24.445 kg	Three Shots .44 Rem. Magnum, Full Copper Alloy Jacket Flat Nose, Soft Core (Lead),Bullet Mass (15.6±0.1 g) Test Range: 5 ±0.5 m. Bullet Velocity (Doppler): 440±10 m/s Striking Distance: 120±10 mm	449.3	445.2	442.3
Status : (P / NS / S) P – Perforation / NS - No Perforation & No Splinters / S - No Perforation with Splinters		NS	NS	NS
Test Results		No Perforation & No Splinters (See Photo No.3)		
Test Piece No.2 40.61 mm Ballistic Glass Batch No.70745 Weight 24.360 kg	Three Shots .44 Rem. Magnum, Full Copper Alloy Jacket Flat Nose, Soft Core (Lead),Bullet Mass (15.6±0.1 g) Test Range: 5 ±0.5 m. Bullet Velocity (Doppler): 440±10 m/s Striking Distance: 120±10 mm	441.4	443.1	448.9
Status : (P / NS / S) P – Perforation / NS - No Perforation & No Splinters / S - No Perforation with Splinters		NS	NS	NS
Test Results		No Perforation & No Splinters (See Photo No.3)		
Test Piece No.B3 40.67 mm Ballistic Glass Batch No.70745 Weight 24.360 kg	Three Shots .44 Rem. Magnum, Full Copper Alloy Jacket Flat Nose, Soft Core (Lead),Bullet Mass (15.6±0.1 g) Test Range: 5 ±0.5 m. Bullet Velocity (Doppler): 440±10 m/s Striking Distance: 120±10 mm	442.1	446.7	431.7
Status : (P / NS / S) P – Perforation / NS - No Perforation & No Splinters / S - No Perforation with Splinters		NS	NS	NS
Test Results		No Perforation & No Splinters (See Photo No.3)		
Status (Comply - See Applied Statement of Conformity and Decision Rule - Page 6)				
EN1063 Performance Requirements P - Perforation: Piercing of a test piece by a bullet or by bullet fragments, and/or creation of an opening from the attack face to the rear face. S - No Perforation with Splinters: No perforation of the test piece by the bullet or parts of the bullet. But with perforation of the witness foil by glass splinters from the protected face. NS – No Perforation & No Splinters: No perforation of the test piece by the bullet or parts of the bullet. And no perforation of the witness foil by glass splinters from rear face.				

RANGE EQUIPMENT & CONFIGURATION

THE GUN

The test sample was shot from the NATO Universal Ballistic Breech with the appropriate standard barrel and bullet type to give projectile stability.

VELOCITY MEASUREMENT

The projectile velocity was measured using Doppler Radar System at 2.5m from the Striking Face with uncertainty 0.4 m/s

Applied Statement of Conformity and Decision Rule:

The uncertainties of measuring devices have been taken into account when making Statement of Conformity and Decision Rule.

Sample Test Level	Test Result	Applied Decision Rule	Statement of Conformity
55.80mm BR-5 (NS) Ballistic Glass (NS)/ Batch No.70745 5.56x45mm "SS109"	No Perforation & No Splinters @ Bullet velocity = 958.3m/s	No perforation of the glazing by the bullet or parts of bullet and No perforation of the witness foil when Bullet velocity@7.5m $\geq$ 941.4 m/s.	Comply
40.60mm BR-4 (NS) Ballistic Glass (NS)/ Batch No.70745	No Perforation & No Splinters @ Bullet velocity = 449.3m/s	No perforation of the glazing by the bullet or parts of bullet and No perforation of the witness foil when Bullet velocity@2.5m $\geq$ 431.4 m/s.	Comply

DISTANCE MESUREMENTS

Test range distance was measured by laser measuring device with uncertainty ± 2 mm, striking distances were marked by a digital caliber with uncertainty ± 0.07 mm.

SUPPORT FIXTURE

The test sample was supported by a fixture that permits its position and attitude to be readily adjusted so that it is perpendicular to the line of flight of the bullet at the point of impact.

AIMING

The aiming process has been achieved by using standard laser bullets.

SHOT PLACEMENT

All shots hit the test sample in the required pattern, with the required place.

SPLINTER COLLECTING BOX & WITNESS FOIL

It is fully enclose the space between the test piece and the 0.02 mm thickness standard witness foil with density 54 g/m² which is mounted in the splinter collecting box in a vertical position at a distance of (500 $\pm$ 10) mm behind the test piece according to EN 1063 standard.

Photos No. 1: Test Samples

55.80mm BR-5 (NS) Ballistic Glass
Batch No.70745

Front veiw



Isometrec veiw



40.60mm BR-4 (NS) Ballistic Glass
Batch No.70745

Front veiw



Isometrec veiw



Photos No. 2: Status of 55.80mm BR-5 (NS) Ballistic Glass/ Batch No.70745 after firing 5.56x45mm “SS109” rounds from 10 m.

Test Piece No.1: 55.80 mm Ballistic Glass/ Size 500x500mm/ Batch No.70745/ Weight 33.335 kg (No Perforation & No Splinters)

Impact Side



Rear Side



Test Piece No.2: 55.73 mm Ballistic Glass/ Size 500x500mm/ Batch No.70745/ Weight 33.625 kg (No Perforation & No Splinters)

Impact Side



Rear Side



Test Piece No.3: 55.85 mm Ballistic Glass/ Size 500x500mm/ Batch No.70745/ Weight 33.380 kg (No Perforation & No Splinters)

Impact Side



Rear Side



Photos No. 3: Status of 40.60mm BR-4 (NS) Ballistic Glass / Batch No.70745 after firing .44 Rem Magnum rounds from 5 m.

Test Piece No.1: 40.62 mm Ballistic Glass/ Size 500x500mm/ Batch No.70745/ Weight 24.445 kg (No Perforation & No Splinters)

Impact Side



Rear Side



Test Piece No.2: 40.61 mm Ballistic Glass/ Size 500x500mm/ Batch No.70745/ Weight 24.360 kg (No Perforation & No Splinters)

Impact Side



Rear Side



Test Piece No.3: 40.67 mm Ballistic Glass/ Size 500x500mm/ Batch No.70745/ Weight 24.360 kg (No Perforation & No Splinters)

Impact Side



Rear Side



APPENDIX A: Approved Technical Testing Proposal




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Test and Evaluation Department
Ballistic Testing Facility
Technical Testing Proposal

Contractor	UNIGLASS /KSA			
Requester	JODDB/Sales			
Reference	Test and Evaluation Request No. 1029 dated 02/02/2022			

Test Sample	Test Type	Test Reference	Test Requirements	Test Criteria
BR-6 (NS) Ballistic Glass 500x500mm Qty. 4	Glass Resistance against bullet attack	Glass in building -Security glazing - Testing and classification of resistance against bullet attack EN 1063 BR6 "Accredited Test"	Firing 3 rounds type 7.62x51mm NATO Ball "DM111" on impact side of each test piece. Test Range: 10±0.5 m. Bullet Velocity (Doppler): 830±10 m/s Striking Distance: 120±10 mm	Annex A
BR-5 (NS) Ballistic Glass 500x500mm Qty. 4		Glass in building -Security glazing - Testing and classification of resistance against bullet attack EN 1063 BR5 "Accredited Test"	Firing 3 rounds type 5.56x45 mm SS109 on impact side of each test piece. Test Range: 10±0.5 m. Bullet Velocity (Doppler): 950±10 m/s Striking Distance: 120±10 mm	
BR-4 (NS) Ballistic Glass 500x500mm Qty. 4		Glass in building -Security glazing - Testing and classification of resistance against bullet attack EN 1063 BR4 "Accredited Test"	Firing 3 rounds type .44 Rem. Magnum on impact side of each test piece. Test Range: 5±0.5 m. Bullet Velocity (Doppler): 440±10 m/s Striking Distance: 120±10 mm	

Note: It is necessary to provide JODDB with serial number of each test piece and also mark impact side prior to testing

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Annex A

Test Criteria Statement of Conformity and Decision Rule			
Test Level	Comply	Comply with Conditions	Not Comply
BR-6 (NS) Ballistic Glass 7.62x51mm DM111	No perforation of the glazing by the bullet or parts of bullet and No perforation of the witness foil when Bullet velocity@7.5m ≥ 821.4 m/s.	No perforation of the glazing by the bullet or parts of bullet and No perforation of the witness foil when 820 m/s ≤ Bullet velocity@7.5m < 821.4 m/s	Perforation of the glazing by the bullet or parts of the bullet or Perforation of the witness foil when Bullet velocity@7.5m ≤ 838.60 m/s.
BR-5 (NS) Ballistic Glass 5.56x45 mm SS109	No perforation of the glazing by the bullet or parts of bullet and No perforation of the witness foil when Bullet velocity@7.5m ≥ 941.4 m/s.	No perforation of the glazing by the bullet or parts of bullet and No perforation of the witness foil when 940 m/s ≤ Bullet velocity@7.5m < 941.4 m/s	Perforation of the glazing by the bullet or parts of the bullet or Perforation of the witness foil when Bullet velocity@7.5m ≤ 958.60 m/s
BR-4 (NS) Ballistic Glass .44 Rem. Magnum	No perforation of the glazing by the bullet or parts of bullet and No perforation of the witness foil when Bullet velocity@2.5m ≥ 431.4 m/s.	No perforation of the glazing by the bullet or parts of bullet and No perforation of the witness foil when 430 m/s ≤ Bullet velocity@2.5m < 431.4 m/s	Perforation of the glazing by the bullet or parts of the bullet or Perforation of the witness foil when Bullet velocity@2.5m ≤ 448.60 m/s.

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END OF THE REPORT